SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Error Analysis Task

Course Code:		Course Title:	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	Assessment:	Assessment Tool 3 – Error Analysis Task
Weightage:	35% of CIA	Duration / Marks:	60 mins / 100 marks
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)
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Section / Batch:	2026	Date:	10/10/2026

Code of Conduct
I M. SRILEKHA(1919260007) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: M. SRILEKHA

Instructions
Read each question carefully; answer all four questions (100 in total). Off-Class Assessment Tool.
Each question gives a student's faulty work. Do not solve from scratch: identify every error (cite it by statement, component, iteration or step number), explain its root cause, correct it with justification and apply the corrected concept. An error merely carried forward counts once.
Follow the conventions stated in each question (alphabetical expansion, explored set, goal test on removal from the frontier, tie-breaking) and present tables neatly.
Marks follow the Error Analysis rubric: error identification, root cause analysis, correction and justification, application of concepts, and clarity and presentation.
Upload in GitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation page at the end).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches to AI, Turing Test, eras and milestones	Q1
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	Q1
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	Q1
4. The Nature of Environments	Task-environment properties and their classification	Q1
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	Q1
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	Q2
7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	Q2, Q4
8. Searching for Solutions	Search graph, frontier, explored set, node expansion	Q3
9. Uninformed Search Strategies	Error tracing of BFS, DFS and uniform-cost search	Q3
10. Search with Partial Information	Sensorless problems, belief states, percept update	Q4

Annexure A – Error Analysis Task

Question 1: Foundations of AI, Intelligent Agents and Rationality – Error Analysis of Concept Notes

Topic: Foundations and history of AI, agents and environments, rationality, nature of environments, structure of agents

A junior engineer prepared the revision notes below on the foundations of AI and intelligent agents. The notes contain errors; some statements may be correct. Analyse them statement by statement.

No.	Statement in the notes
S1	The Turing Test (1950) judges a machine to be intelligent if its internal reasoning always follows the laws of logic. It is therefore the standard example of the "thinking rationally" approach to AI.
S2	The "AI winters", periods of reduced funding and interest, occurred in the 1950s, before the 1956 Dartmouth workshop. That workshop then revived the field and coined the term "artificial intelligence".
S3	A rational agent always achieves the best possible outcome because it knows the actual result of every action in advance. Hence a taxi that picks a route later blocked by an unforeseeable landslide was not rational.

No.	Statement in the notes
S4	PEAS description of an automated taxi – Performance measure: steering, accelerator, brake, horn. Environment: roads, other traffic, pedestrians, weather. Actuators: safe, fast, legal and comfortable trip. Sensors: cameras, GPS, speedometer.
S5	Chess played without a clock is a partially observable, stochastic, episodic, static and discrete task environment.
S6	A model-based reflex agent keeps an internal state, updated using knowledge of how the world evolves and how its own actions affect it. It can therefore act sensibly even when the current percept does not reveal the full state of the world.

(a) Identify which statements are correct and which are incorrect. For each incorrect statement, quote the erroneous phrase or claim (one statement contains more than one error). CSA1716

Statement	Correct / Incorrect	Erroneous phrase / claim
S1	Incorrect	The Turing Test judges intelligence by conversation, not by whether internal reasoning always follows logic.
S2	Incorrect	AI winters did not occur in the 1950s before Dartmouth; they occurred later. Dartmouth (1956) helped establish the field and the name AI.
S3	Incorrect	“always achieves the best possible outcome”; “knows the actual result ... in advance”; landslide example is wrongly called irrational.
S4	Incorrect	Performance measure and actuators are swapped: steering/accelerator/brake/horn are actuators; safe/fast/legal/comfortable are performance criteria.
S5	Incorrect	Chess without a clock is not partially observable, stochastic, or episodic; it is fully observable, deterministic, sequential, static, discrete.
S6	Correct	No erroneous phrase. The statement accurately describes a model-based reflex agent.

(b) Root cause analysis: for every error, explain the underlying misconception, i.e. which concept has been misunderstood and why the statement goes wrong.

Error (statement / part)	Root cause (misconception)
S1: Turing Test	Confuses human-like behavior (acting humanly) with logical reasoning (thinking rationally).
S2: AI winters	Misplaces the AI winters historically; the Dartmouth workshop was in 1956 and introduced the term AI.
S3: Rationality	Confuses rationality with omniscience. Rationality depends on percept history, available actions, knowledge and performance measure, not guaranteed success.
S4: PEAS	Confuses actuators (means of acting) with performance measures (criteria for success).
S5: Chess environment	Misclassifies observability, determinism, episodic/sequential, and static/dynamic properties.

(c) Rewrite each faulty statement correctly and justify your correction. For a statement that is already correct, write “No change” and justify why.

Statement	Corrected statement	Justification
S1	The Turing Test evaluates whether a machine’s behavior in conversation is indistinguishable from a human’s; it is an example of the acting humanly approach.	It tests observable behavior, not whether internal reasoning follows formal logic.
S2	AI winters were periods of reduced funding and interest that occurred later, notably in the 1970s and again in the late 1980s. The 1956 Dartmouth workshop helped establish AI as a field and coined the term.	The dates and historical sequence are corrected.
S3	A rational agent selects the action expected to maximize its performance measure given its percept sequence and available knowledge. The taxi can be rational even if an unforeseeable landslide later blocks its route.	Rationality is judged using information available at decision time, not perfect foresight.

Statement	Corrected statement	Justification
S4	Performance measure: safe, fast, legal, comfortable trip. Environment: roads, traffic, pedestrians, weather. Actuators: steering, accelerator, brake, horn. Sensors: cameras, GPS, speedometer.	PEAS correctly separates success criteria, world, action mechanisms and percept mechanisms.
S5	Chess without a clock is fully observable, deterministic, sequential, static and discrete.	The board and pieces are observable; moves have known outcomes; each move affects later moves; no clock means the environment does not change while a player thinks.
S6	No change: a model-based reflex agent maintains internal state updated from percepts and knowledge of world/action effects, allowing it to act with incomplete current percepts.	This is the defining feature of a model-based reflex agent.

(d) Apply the corrected concepts to the following case. A self-driving taxi chooses a highway route that every available traffic report showed to be clear; an unforeseeable landslide then blocks the road.

(i) Using the four factors that decide whether an agent is rational, argue whether the taxi's choice was rational.

The four factors are:

Performance measure: The taxi aims to provide a safe, fast, legal and comfortable journey.

Percept sequence: All available traffic reports showed that the highway was clear.

Prior knowledge: The taxi used the available information and knowledge about the road conditions.

Available actions: The taxi selected the highway route from the available route options.

Conclusion: The taxi's choice was rational because it selected the best route based on the available information. The landslide was unforeseeable, so the taxi cannot be considered irrational just because the road became blocked.

(ii) Name the properties of the taxi's task environment that make perfect foresight impossible.

Partially observable: The taxi cannot know every road condition.

Stochastic: Unpredictable events, such as landslides, can occur.

Dynamic: Road conditions can change while the taxi is travelling.

Sequential: Each action affects the next decision and the overall journey.

(iii) Describe one situation in which a simple reflex taxi would behave poorly, and state what a model-based reflex taxi keeps in order to avoid it.

(i)	A simple reflex taxi may behave poorly when it encounters a road blockage because it reacts only to the current percept and does not remember past information.
(ii)	Yes. Rationality is judged by the percept sequence, available actions, knowledge and performance measure. Reports showed the highway clear, so choosing it was reasonable; an unforeseeable landslide does not by itself make the choice irrational.
(iii)	The environment is partially observable, stochastic/uncertain and dynamic; it is also sequential. These properties mean the taxi cannot know every future event in advance.

Question 2: Problem-Solving Agents and Problem Formulation – Error Analysis of a Formulation

Topic: Problem-solving agents, problem formulation

A student formulates the Missionaries and Cannibals problem for a problem-solving agent. Three missionaries and three cannibals stand on the left bank of a river, with a boat that carries one or two people and cannot cross empty. The boat is on the left bank at the start. Missionaries must never be outnumbered by cannibals on either bank (a bank with no missionaries is safe). The objective is to move everyone to the right bank using the fewest crossings.

Component	Student's formulation
State representation	(M, C) = number of missionaries and cannibals on the left bank
Initial state	(3, 3)
Actions	Take the boat across with 1M, 2M, 1C, 2C or 1M + 1C
Goal test	M = C
Legality check	A state is legal if C ≤ M on the left bank (or M = 0 there)
Path cost	Number of people carried in the crossing
Note N1	The agent works in the order: search → goal formulation → problem formulation → execution.

Component	Student's formulation
Note N2	The state space contains 16 states, of which 10 are legal.

(a) For each component and note, state whether it is correct or incorrect; for each incorrect one, state what is wrong.

Component / Note	Correct / Incorrect	What is wrong
State representation	Incorrect	Boat side is missing from the state.
Initial state	Incorrect	Boat location is missing from the initial state.
Actions	Correct	All five listed loads obey the boat capacity and non-empty crossing rule.
Goal test	Incorrect	M=C does not mean everyone has crossed; the goal is M=0 and C=0 on the left.
Legality check	Incorrect	Only the left bank is checked; safety must be checked on both banks.
Path cost	Incorrect	Cost should count crossings, not people carried.
Note N1	Incorrect	Goal formulation and problem formulation should come before search and execution.
Note N2	Incorrect	Including boat side gives 20 legal states, not 16 states with 10 legal.

(b) Root cause analysis: for every error, explain the underlying cause, i.e. which principle of problem formulation or of the problem-solving agent has been violated.

Error (component / note)	Root cause
State representation	The state must include all information needed to determine legal actions, including the boat side.
Initial state	The initial state must specify the boat on the left.
Goal test	A goal test must express the actual objective: everyone on the right bank.
Legality check	The missionaries/cannibals safety constraint applies to both banks.
Path cost	The stated objective is fewest crossings, so each crossing has unit cost.
Note N1	A problem-solving agent formulates a goal, formulates the problem, searches, and then executes.
Note N2	There are 10 safe (M,C) pairs; with two boat positions, there are 20 legal states.

(c) Give the corrected version of every faulty component or note and justify each correction.

Component / Note	Corrected formulation	Justification
State representation	(M,C,B), where M,C are the numbers on the left and B is L or R.	Boat side is necessary to determine which crossing actions are possible.
Initial state	(3,3,L).	Everyone and the boat begin on the left.
Goal test	M=0 and C=0 on the left, with the boat on the right at completion.	This means all people have reached the right bank.
Legality check	Left safe if M=0 or M≥C; right safe if (3–M)=0 or (3–M)≥(3–C).	Both banks must satisfy the rule.
Path cost	Cost 1 for each crossing; minimize the total number of crossings.	This matches the objective.
Note N1	Goal formulation → problem formulation → search → execution.	This is the correct agent sequence.
Note N2	There are 10 safe (M,C) pairs; multiplying by 2 boat positions gives 20 legal states.	Boat side is part of the state.

(d) Apply your corrected formulation.

(i) Complete the table of boat loads available from the initial state, writing each resulting state as (M, C, boat side) and whether it is legal, with the reason.

Boat load	Resulting state (M, C, boat)	Legal? Reason
1M	(2,3,R)	Illegal: left bank has 2 missionaries and 3 cannibals, so missionaries are outnumbered.
2M	(1,3,R)	Illegal: left bank has 1 missionary and 3 cannibals.
1C	(3,2,R)	Legal: left bank has 3M/2C; right bank has no missionaries, so safe.

Boat load	Resulting state (M, C, boat)	Legal? Reason
2C	(3,1,R)	Legal: left bank has 3M/1C; right bank has no missionaries, so safe.
1M + 1C	(2,2,R)	Legal: left bank has 2M/2C and right bank has 1M/1C.

(ii) Give one state that the student's legality check accepts although it is actually illegal.

State: (M, C, Boat) = (2, 3, R)

Reason: The student's check considers only the left bank. Since is false (3 > 2), this particular state is rejected by the student's check, so it is not a valid example.

A correct example is (1, 2, R).

Student's check: Accepts it because on the left bank, is false (2 > 1), so it would actually reject it too.

Better example: (0, 3, R) is accepted because , but it is actually illegal if the right bank has 3 missionaries and 0 cannibals? No—the right bank is safe.

(iii) Using your corrected path cost, state the cost of an optimal solution.

(ii)	(ii) Example: (M,C)=(2,0). The student's check accepts it because 0≤2, but the right bank has 1 missionary and 3 cannibals, so it is illegal.
(iii)	(iii) Optimal path cost = 11 crossings, because each crossing costs 1 and the objective is to minimize crossings.

Question 3: Searching for Solutions and Uninformed Search – Error Analysis of Search Traces

Topic: Searching for solutions, BFS, DFS, uniform-cost search

A student ran BFS, DFS and uniform-cost search (UCS) on the weighted undirected graph below. The start node is S and the goal node is G. Conventions: graph search with an explored set; neighbours expanded alphabetically; goal test when a node leaves the frontier; ties broken alphabetically. BFS and DFS ignore edge costs while searching, but path costs are measured with them. BFS: do not add a node that is already explored or already in the queue. DFS: use a stack with its top on the left; push children so that the alphabetically first child is popped first; do not push a node that is already explored or already on the stack. UCS: write entries as Node:g(parent) and replace an entry whenever a cheaper path to a frontier node is found.

Edge	Cost	Edge	Cost
S – C	3	B – G	3
S – D	3	D – E	2
A – B	2	D – G	6
A – C	6	E – G	3
B – D	4		

Student's BFS trace (queue initially [S])

Iteration	Node Dequeued	Queue after Expansion	Explored Set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[B, E, G, A]	{S, C, D}
4	B	[E, G, A]	{S, C, D, B}
5	E	[G, A]	{S, C, D, B, E}
6	G (goal)	[A]	{S, C, D, B, E}

BFS result: path S – D – G, cost 9, 6 nodes removed from the frontier.

Student's DFS trace (stack initially [S], top on the left)

Iteration	Node Popped	Stack after Pushing Children
1	S	[D, C]
2	D	[G, E, B, C]
3	G (goal)	[E, B, C]

DFS result: path S – D – G, cost 9, 3 nodes removed from the frontier.

Student's UCS trace

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), A:9(C), G:9(D)
5	B	7	A:9(C), G:9(D)
6	A	9	G:9(D)
7	G (goal)	9	–

UCS result: path S – D – G, cost 9, 7 nodes removed from the frontier.

Student's remarks

No.	Remark
R1	An explored set is not needed in BFS because BFS never revisits a node, so the Explored Set column is unnecessary.
R2	All three strategies returned S – D – G with cost 9, so all three are optimal and the traces are confirmed correct.
R3	DFS removed the fewest nodes from the frontier (3), so DFS is the best strategy for this graph.

(a) Examine the three traces and the remarks. List every error, giving the trace and iteration (or the remark number) and what is wrong. An error that is only carried forward counts once.

No.	Where (trace / iteration / remark)	What is wrong
1	BFS, iteration 3	Queue should be [A, B, E, G]; A was already in the queue and must remain first, while new neighbours B, E, G are appended alphabetically.
2	BFS, iterations 4–6/result	Because the queue at iteration 3 is wrong, the dequeue order/path and node count are wrong. Correct order is S,C,D,A,B,E,G.
3	DFS, iteration 1	Children must be pushed so the alphabetically first is popped first; stack should be [C,D], not [D,C].
4	DFS, later trace/result	The wrong first stack order causes the wrong path. Correct DFS visits S,C,A,B,G and costs 14.
5	UCS, iteration 4	From E at cost 5, G can be reached at cost 8, cheaper than the queued G:9(D); replace it with G:8(E).
6	UCS, later trace/result	The cheaper G entry must be removed first; UCS solution is S-D-E-G with cost 8, not S-D-G with cost 9.
7	R1–R3	R1 is false: graph search needs an explored set. R2 is false because BFS/DFS are not cost-optimal here and traces are wrong. R3 incorrectly treats fewest removals as the sole measure of best strategy.

(b) Root cause analysis: for each error listed in (a), explain the rule or concept that was violated and how it led to the wrong entry or conclusion.

No.	Root cause (rule or concept violated)
1	BFS must preserve FIFO order and not add a node already in the queue.
2	BFS finds the shallowest path by edge count; the trace must follow the corrected queue.
3	Because a stack is LIFO, reverse-push children to pop the alphabetically first child first.
4	DFS follows stack order; its first goal path need not be shortest or cheapest.
5	UCS updates a frontier node whenever a cheaper path is found.
6	UCS expands the lowest cumulative-cost entry; G at cost 8 precedes A at 9.
7	Explored sets prevent repeated expansion; optimality depends on strategy and path-cost assumptions.

(c) Write the corrected traces and justify each correction.

(i) BFS: the corrected queue after iteration 3 and the full order in which nodes are dequeued.

Queue after iteration 3	[A, B, E, G]
Order of dequeued nodes	S, C, D, A, B, E, G

(ii) DFS: the corrected trace.

Iteration	Node Popped	Stack after Pushing Children
1	S	[C, D]
2	C	[A, D]
3	A	[B, D]
4	B	[G, D]
5	G	Goal

(iii) UCS: the corrected trace.

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), G:8(E), A:9(C)
5	B	7	G:8(E), A:9(C)
6	G	8	A:9(C)

Justification of the corrections	BFS keeps FIFO order and avoids duplicate frontier nodes. DFS uses a LIFO stack with reverse-push ordering. UCS replaces a frontier entry when a cheaper path is found and expands the lowest-cost entry; G is reached at cost 8 through E.

(d) Apply the corrected traces.

(i) Complete the comparison table.

Strategy	Path Found	Path Cost	Nodes Removed from Frontier	Optimal Here?
BFS	S-C-A-B-G	14	7	No
DFS	S-C-A-B-G	14	5	No
UCS	S-D-E-G	8	6	Yes

(ii) Use iteration 3 of the corrected BFS trace to explain why an explored set is needed in graph search.

(ii)	At iteration 3, D generates B, E and G, while A is already in the frontier. Frontier and explored checks prevent duplicates and redundant expansions.

Question 4: Search with Partial Information – Error Analysis of Belief-State Computation

Topic: Problem formulation, search with partial information (sensorless and percept-based belief-state search)

A vacuum agent lives in a two-square world (Left square, Right square), each square being Clean (C) or Dirty (D). Actions: Left, Right and Suck (Left in the left square and Right in the right square leave the agent where it is; Suck cleans the current square). The goal is that both squares are Clean. Physical states are written (Agent, Left square, Right square):

State	Description	State	Description
S1	(L, D, D)	S5	(L, C, D)
S2	(R, D, D)	S6	(R, C, D)
S3	(L, D, C)	S7	(L, C, C)
S4	(R, D, C)	S8	(R, C, C)

Task A – sensorless agent. The agent knows it is in the Left square but nothing about the dirt, so its initial belief state is {S1, S3, S5, S7}. A student applied the plan [Suck, Right, Suck]:

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}

Step	Action	Belief State after Action
3	Suck	{S6, S8}

Task B – agent with a local sensor that reports [location, status of the current square]. Starting from the belief state of all eight states, the agent receives the percept [Left, Dirty], performs Suck, receives [Left, Clean], performs Right and receives [Right, Dirty]. The student's table:

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S2, S3, S6}
1	Suck	{S4, S5, S7, S8}	[Left, Clean]	{S5, S7}
2	Right	{S6, S7}	[Right, Dirty]	{S6}

Student's conclusions

No.	Conclusion
C1	Task A: the final belief state still contains two states, so the agent can never know where it is. No plan can guarantee the goal when the agent has no sensors; hence sensorless problems cannot be solved.
C2	Task B: the belief state is now the singleton {S6}, so the agent knows its location and the dirt status, and the goal has been reached.
C3	With 8 physical states there are only 8 belief states (one per physical state), so searching over belief states is no harder than searching over physical states.

(a) Examine the tables and the conclusions. List every error, giving the task and step (or the conclusion number) and what is wrong. An error that is only carried forward from an earlier wrong row counts once.

No.	Where (task / step / conclusion)	What is wrong
1	Task A, step 3	Final Suck changes S6=(R,C,D) to S8=(R,C,C); S8 remains S8. Final belief should be {S8}, not {S6,S8}.
2	Task B, step 0 percept update	[Left, Dirty] keeps only states with agent at L and left square D: {S1,S3}, not {S1,S2,S3,S6}.
3	Task B, step 1 predicted action	Suck applied to {S1,S3} yields {S5,S7}; the student predicted from an incorrect belief state.
4	Task B, step 2 predicted action	Right from {S5,S7} yields {S6,S8}, not {S6,S7}.
5	Task B, step 2 percept update	[Right, Dirty] filters {S6,S8} to {S6}; this is the correct update once the prediction is corrected.
6	Conclusion C1	A sensorless plan can guarantee the goal even if location/dirt are initially uncertain; Task A ends at {S8}, where both squares are clean.
7	Conclusions C2/C3	{S6} is not a goal because the right square is dirty. Eight physical states yield 2^8=256 possible belief states, not 8.

(b) Root cause analysis: for each error listed in (a), explain the concept or rule of belief-state search that was misapplied.

No.	Root cause (concept or rule misapplied)
1	Belief states are transformed by applying the action to every possible physical state; the final Suck cleans the right square.
2	A percept update filters states by both reported location and the current square's dirt status.
3	Action prediction must be computed from the corrected preceding belief state.
4	Right changes the agent location in every possible state; it does not change dirt status.
5	Percepts eliminate states inconsistent with the observation.
6	Sensorless agents can achieve a goal if a plan maps every possible state in the belief state to a goal state.
7	A belief state is any subset of physical states, so n physical states permit 2^n subsets; a singleton does not imply the goal is satisfied.

(c) Write the corrected tables and conclusions, with justification.

(i) Task A: the corrected belief states.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S8}

(ii) Task B: the corrected belief states after each action and each percept.

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S3}
1	Suck	{S5, S7}	[Left, Clean]	{S5, S7}
2	Right	{S6, S8}	[Right, Dirty]	{S6}

(iii) Corrected conclusions C1 and C2, with justification.

C1	C1 is incorrect. Task A ends in {S8}, which represents (R,C,C), so both squares are clean. A sensorless agent can solve a problem without identifying one exact physical state at every step.
C2	C2 is incorrect. {S6} means (R,C,D): the agent is on the right, the left square is clean, and the right square is dirty. It knows the state but has not reached the goal.

(d) Apply the corrected concepts.

(i) How many different belief states are possible when there are 8 physical states? Why does this make belief-state search harder than search over physical states?

(i)	There are 2^8 = 256 possible belief states, because each belief state is a subset of the 8 physical states. This is much larger than 8 physical states, so belief-state search can have an exponentially larger state space.

(ii) For the initial belief state {S1, S3, S5, S7}, apply the plan [Right, Suck], write the belief state after each action, and state whether it is a solution.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Right	{S2, S4, S6, S8}
2	Suck	{S4, S8}
	Solution? Why?	Not a solution: S4=(R,D,C) still has the left square dirty.

Rubrics for Error Analysis

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Error Identification	20	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors
Root Cause Analysis	30	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	25	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified	Incorrect or no corrections provided
Application of Concepts	15	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	10	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria	Error Identification	Root Cause Analysis	Correction & Justification	Application of Concepts	Clarity & Presentation	Sub. Total	Total %
1							
2							

Q. No. / Criteria	Error Identification	Root Cause Analysis	Correction & Justification	Application of Concepts	Clarity & Presentation	Sub. Total	Total %
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____